Heating of planetesimals from ^{60}Fe & ^{26}Al

Hunting the source of short-lived radioisotopes and simulating desiccation in planetesimals

Dr. Joseph Eatson - University of Sheffield
Supervisor: Dr. Richard Parker - University of Sheffield

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[1] Fu & Elkins-Tanton 2014

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How do disks survive this enrichment?

How do they affect planet formation?

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- Primary heating source in the early solar system [1].
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- Wolf-Rayet (WR) winds and supernovae are sources of SLRs.

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Planetesimal desiccation

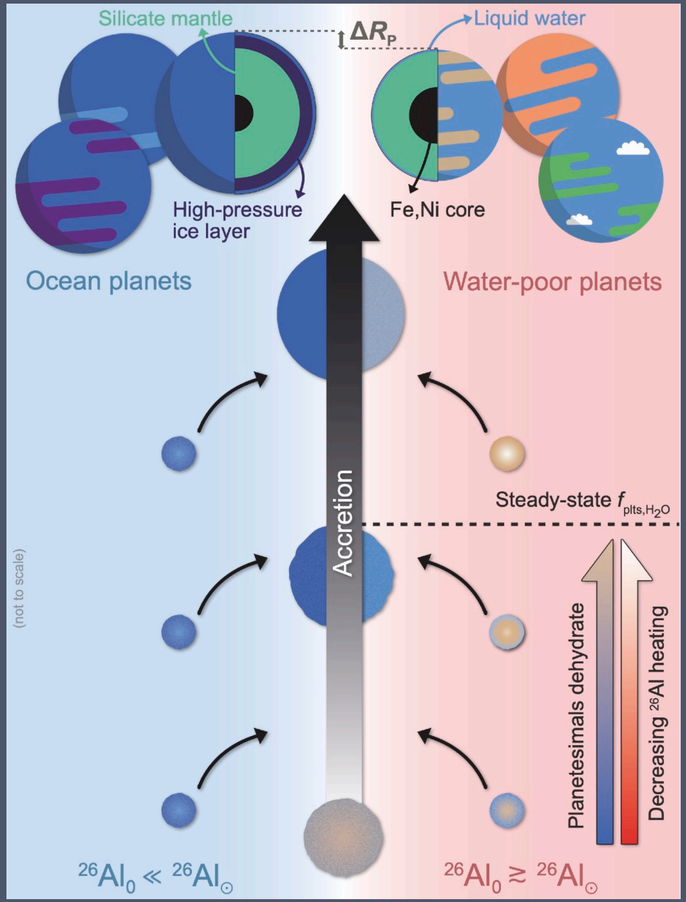
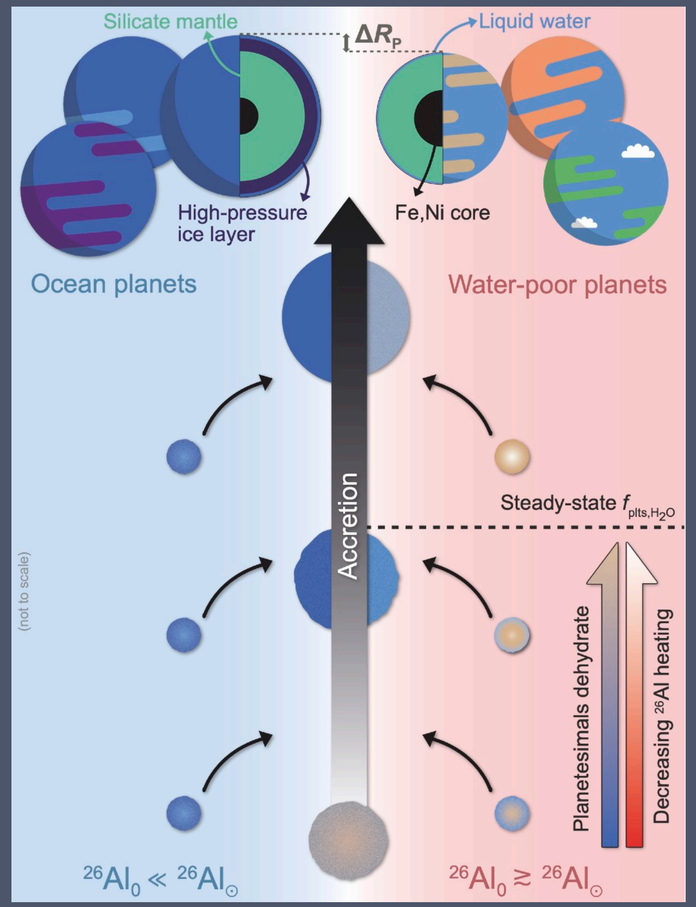


Fig. Lichtenberg+ 2019, [1] Lichtenberg+ 2019

Planetesimal desiccation

- Heating = vaporisation and outgassing [1].



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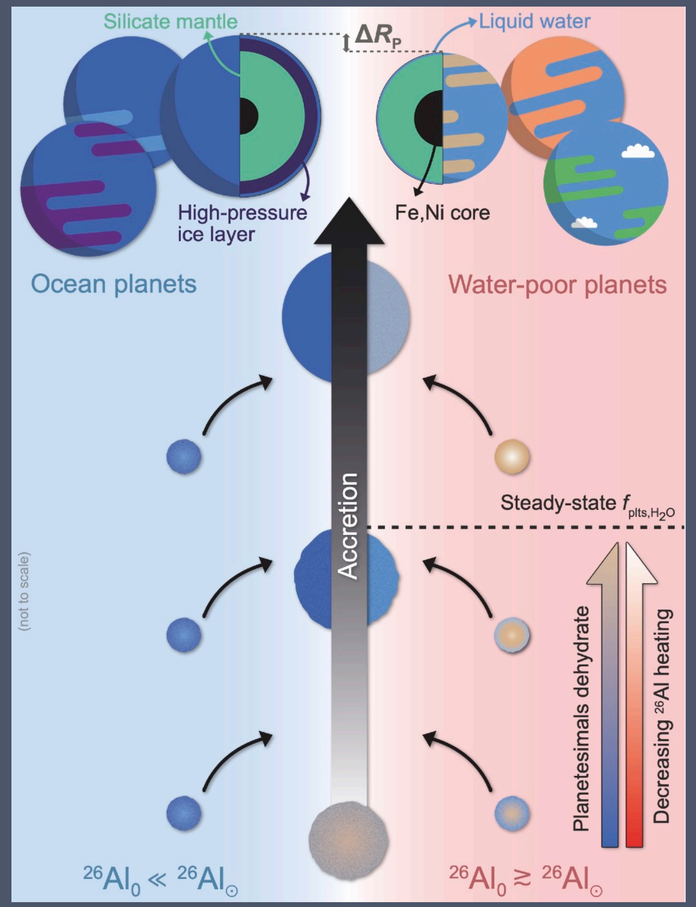

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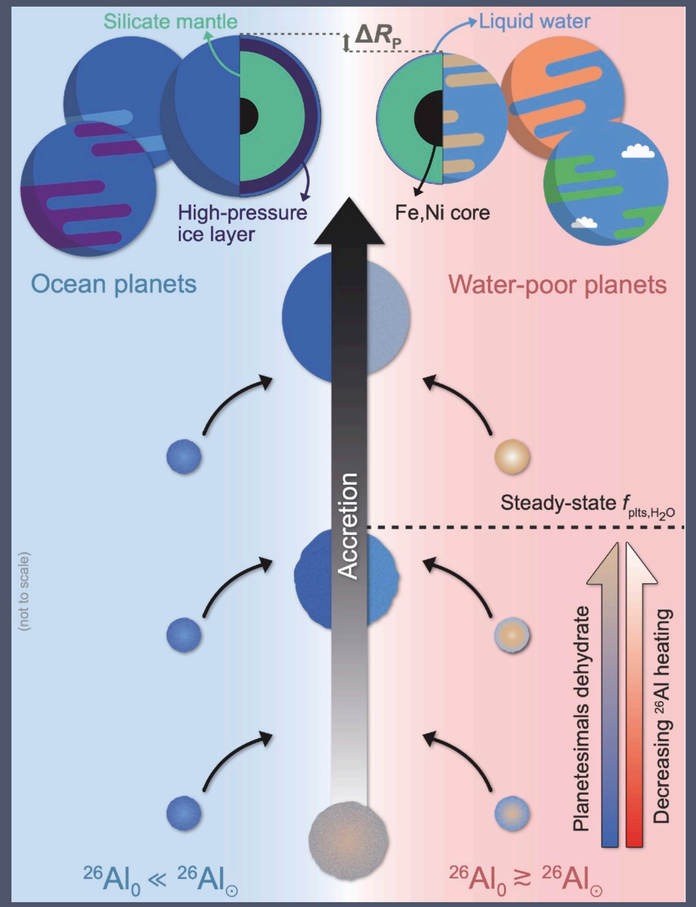

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SLR enrichment of disks

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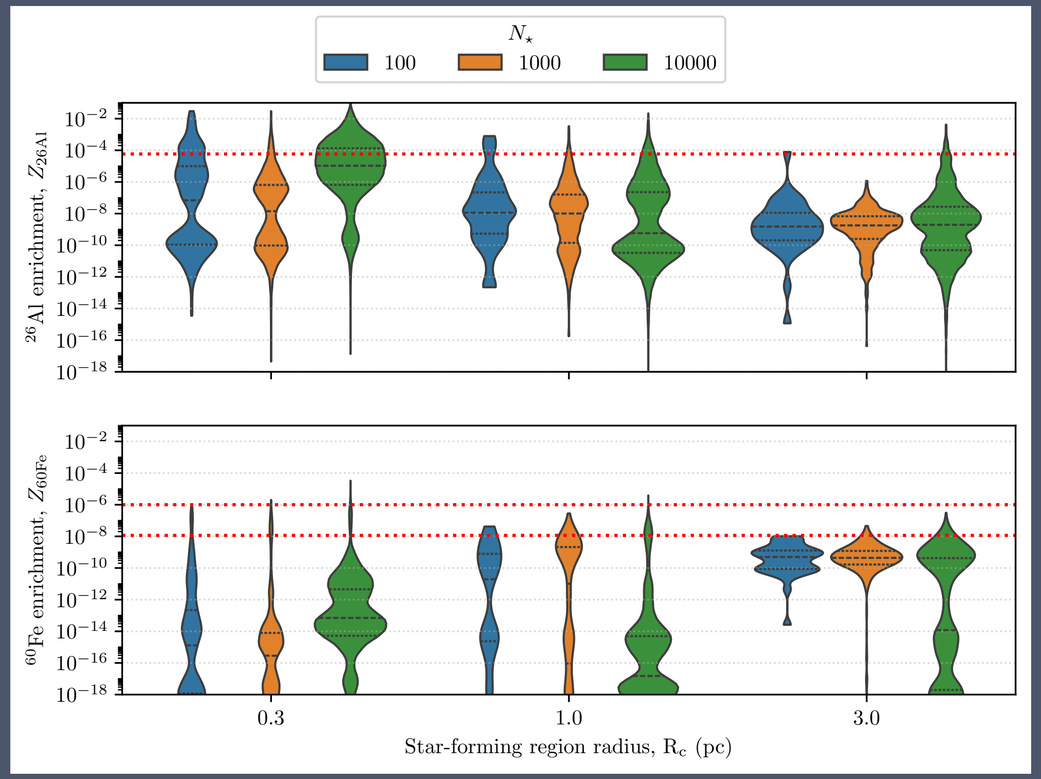
Is there another pre-formation mechanism?

Enriching disks

Eatson, Parker & Lichtenberg 2024



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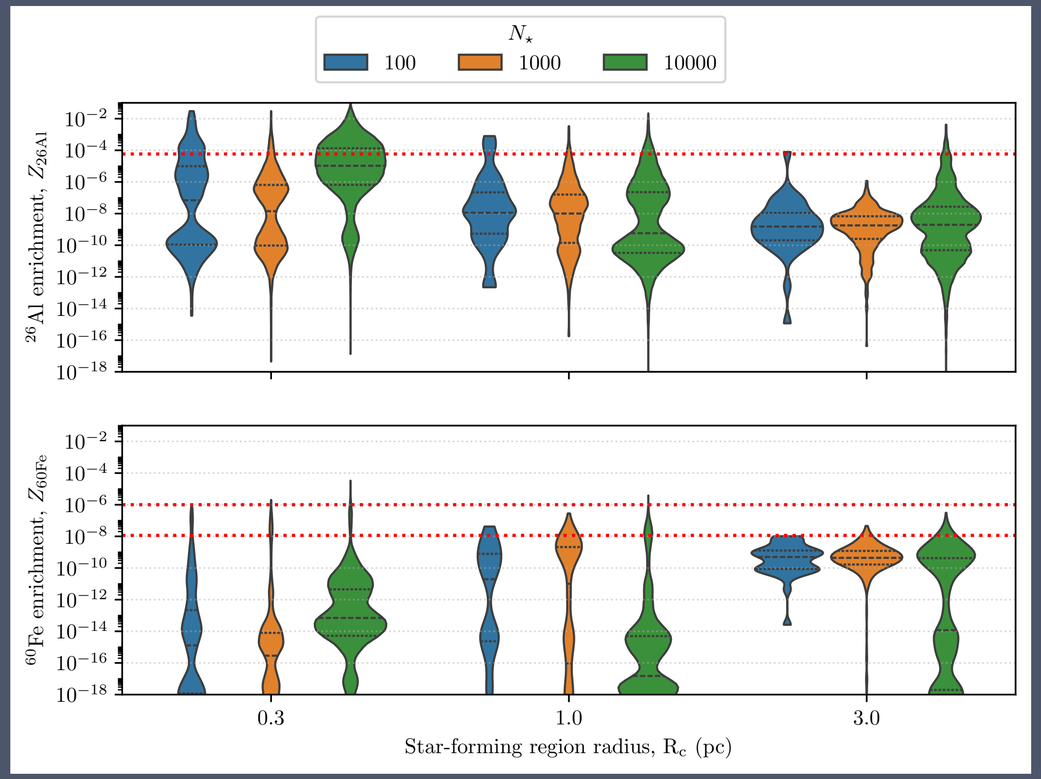

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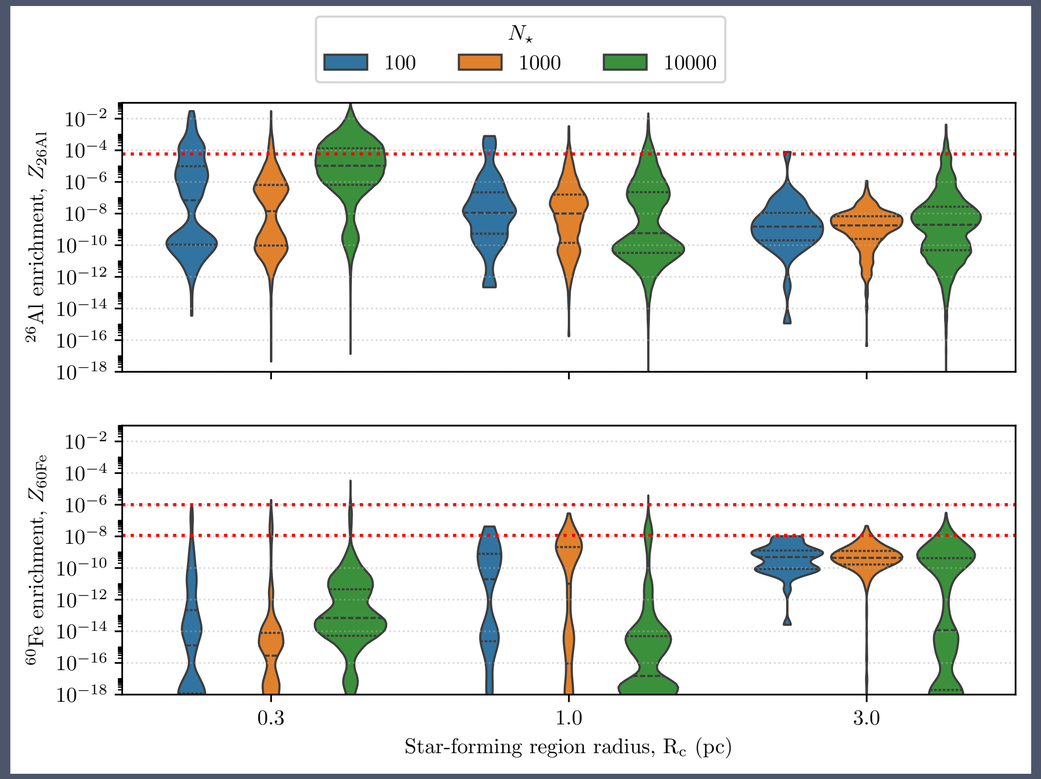

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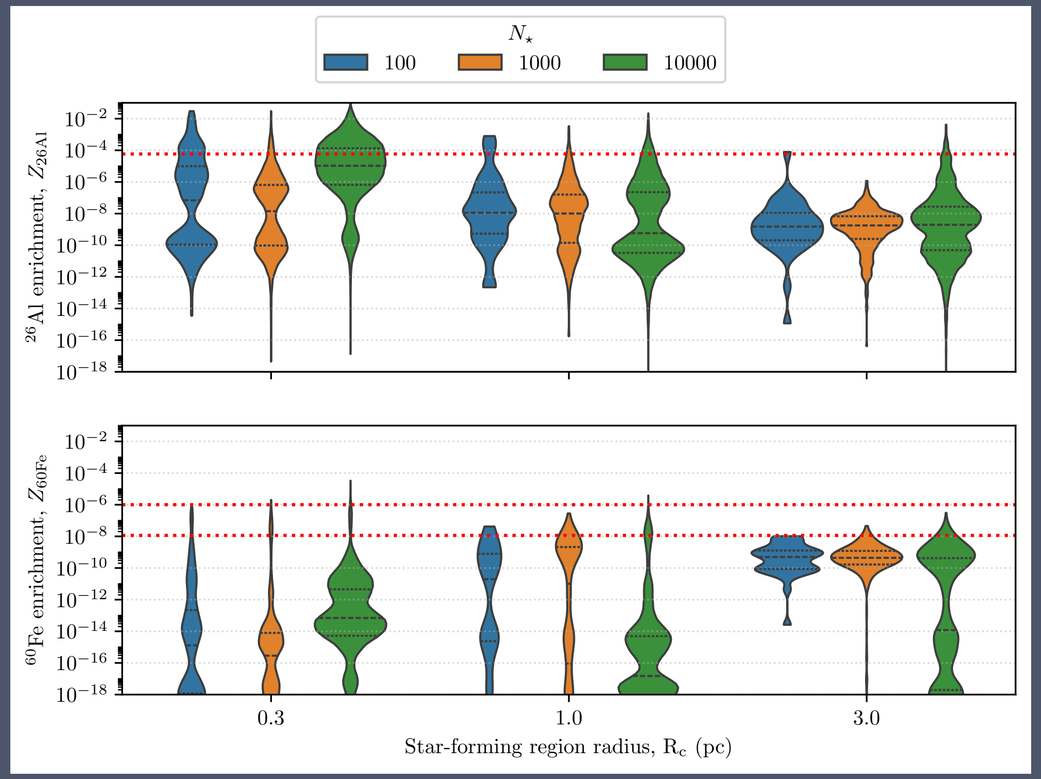

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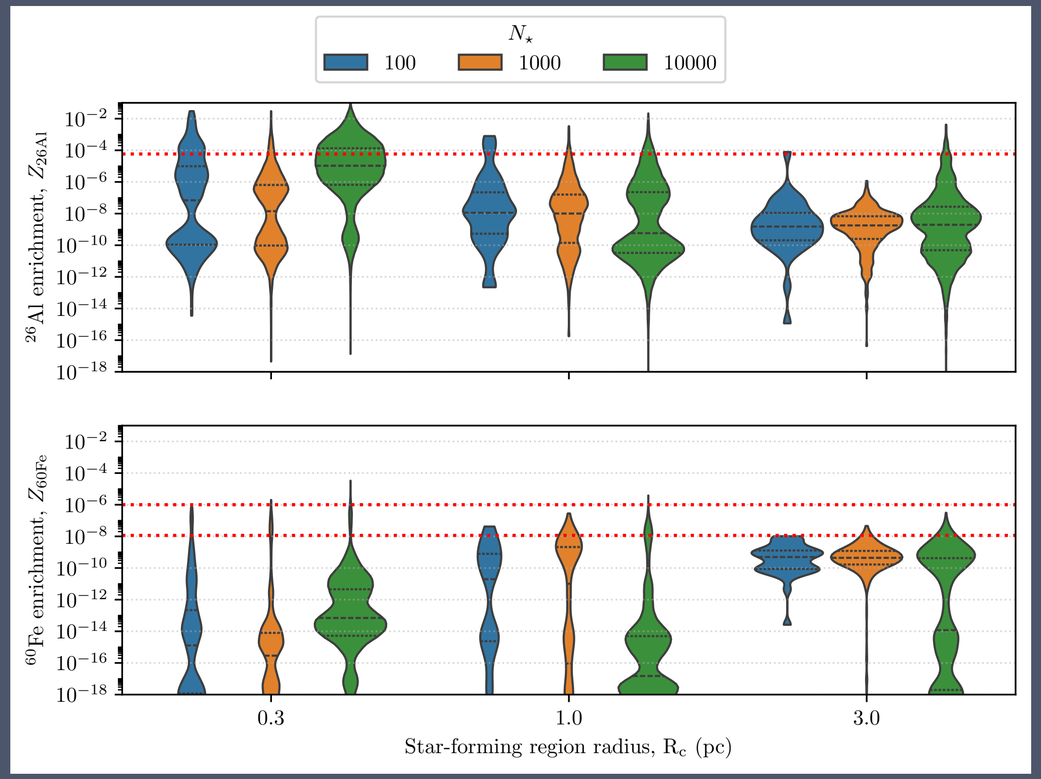

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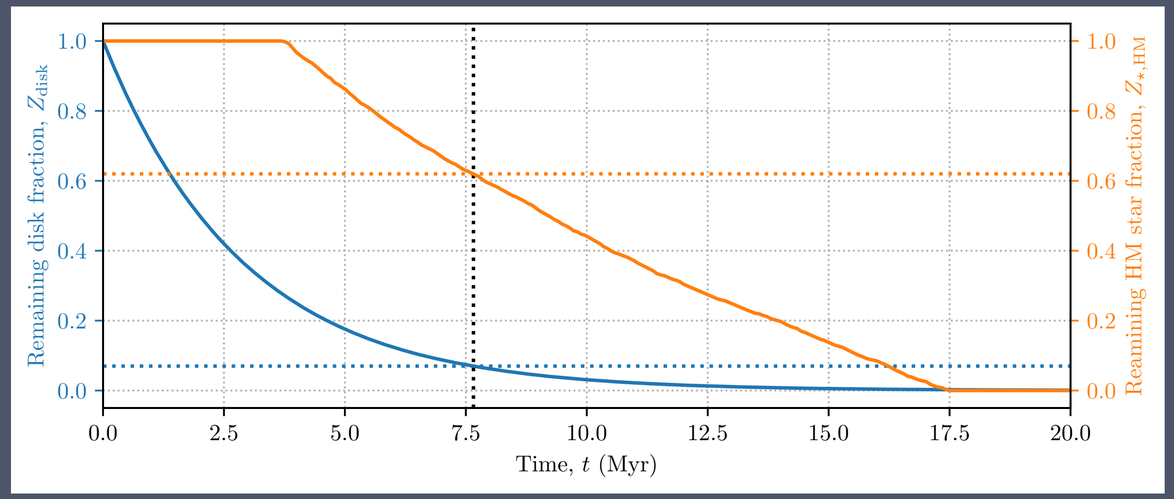
The hitch with winds and SNe



X-axis: time
Y-axis left/blue: Remaining disks
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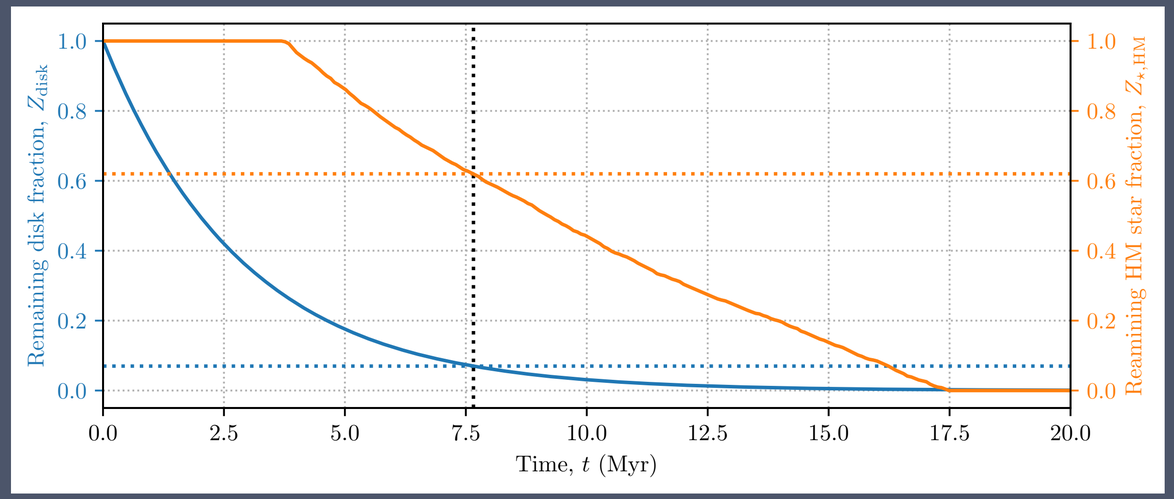

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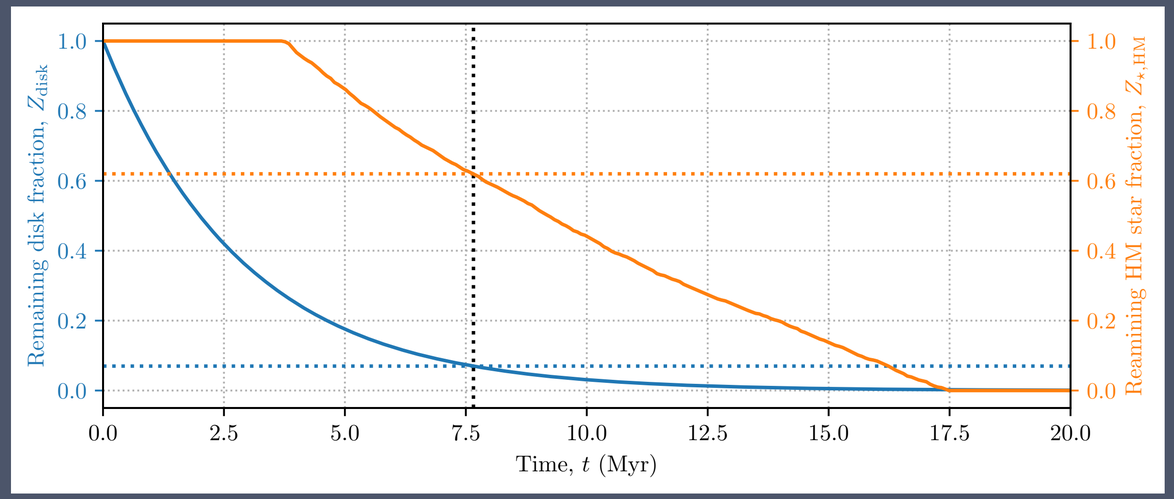

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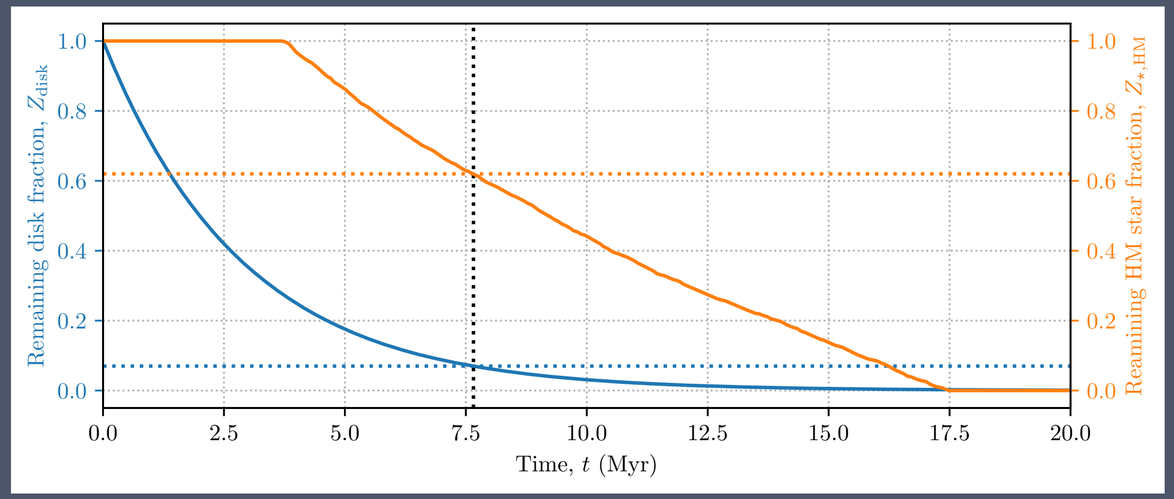

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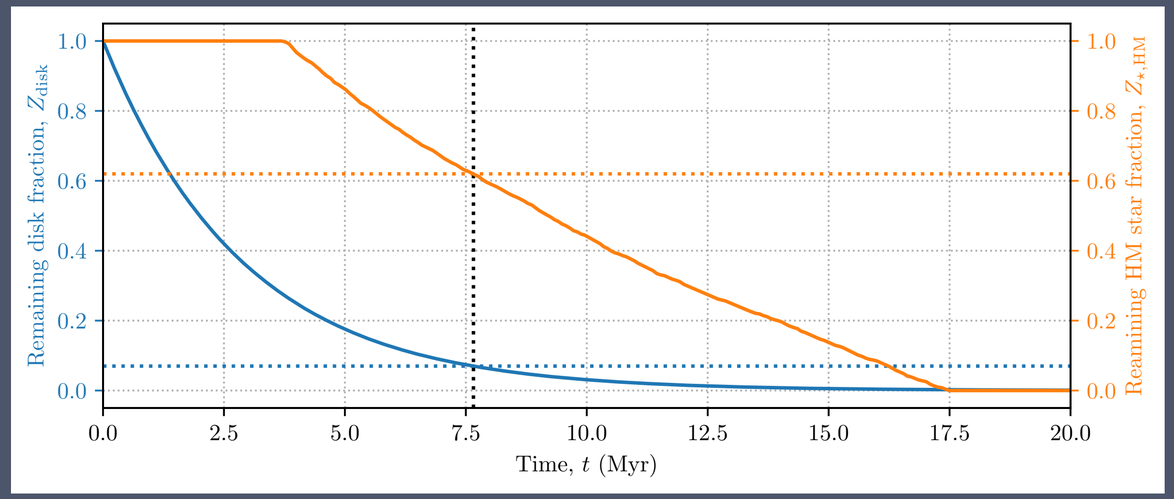
- Other enrichment methods?



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[1] Parker & Schoettler, 2023

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Interlopers?

- AGB star observed to pass through cluster [1].
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- Gentler winds, less UV flux.

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AGBI sensitivity to...

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Interloper evolution?

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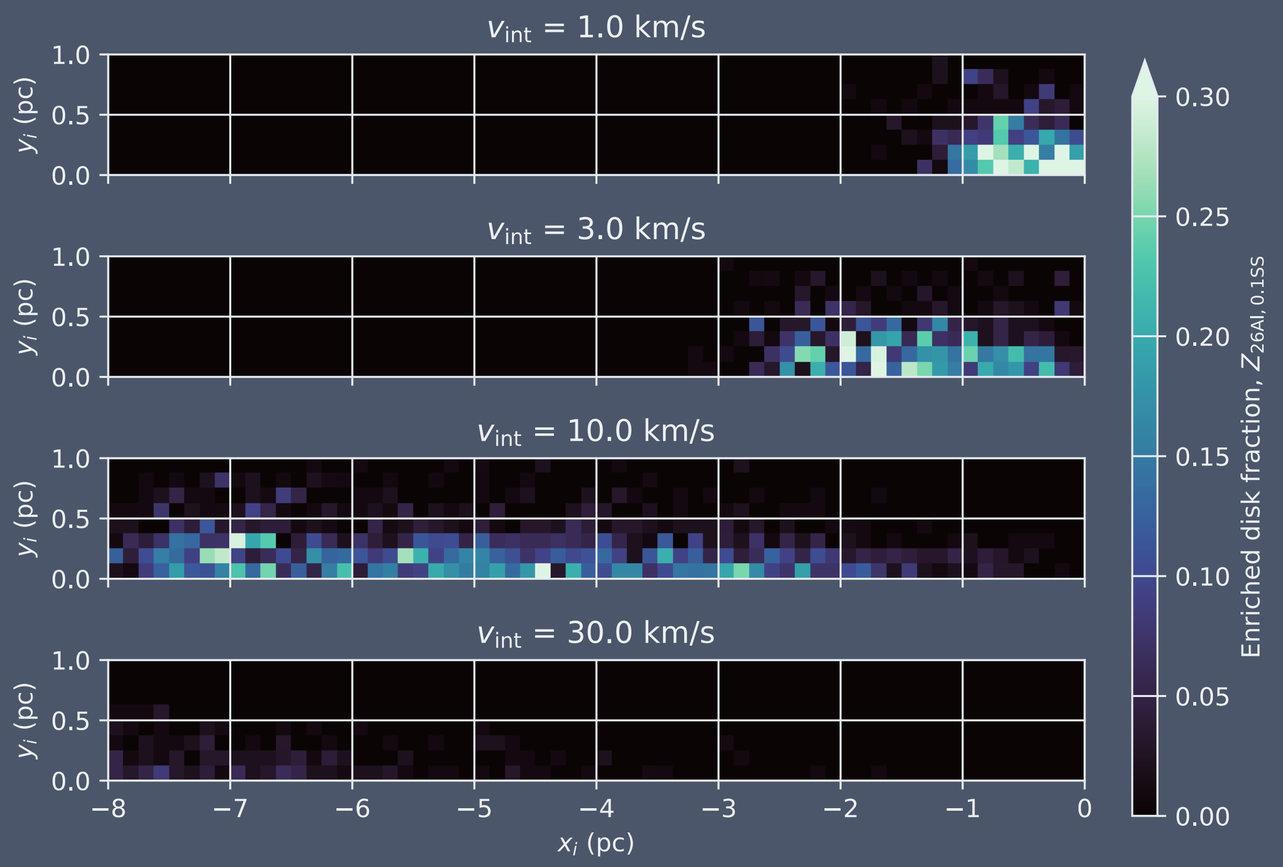
Interloper evolution?

Encounter velocity?

Interlopers!

Eatson & Parker
In prep.
QR code for poster!



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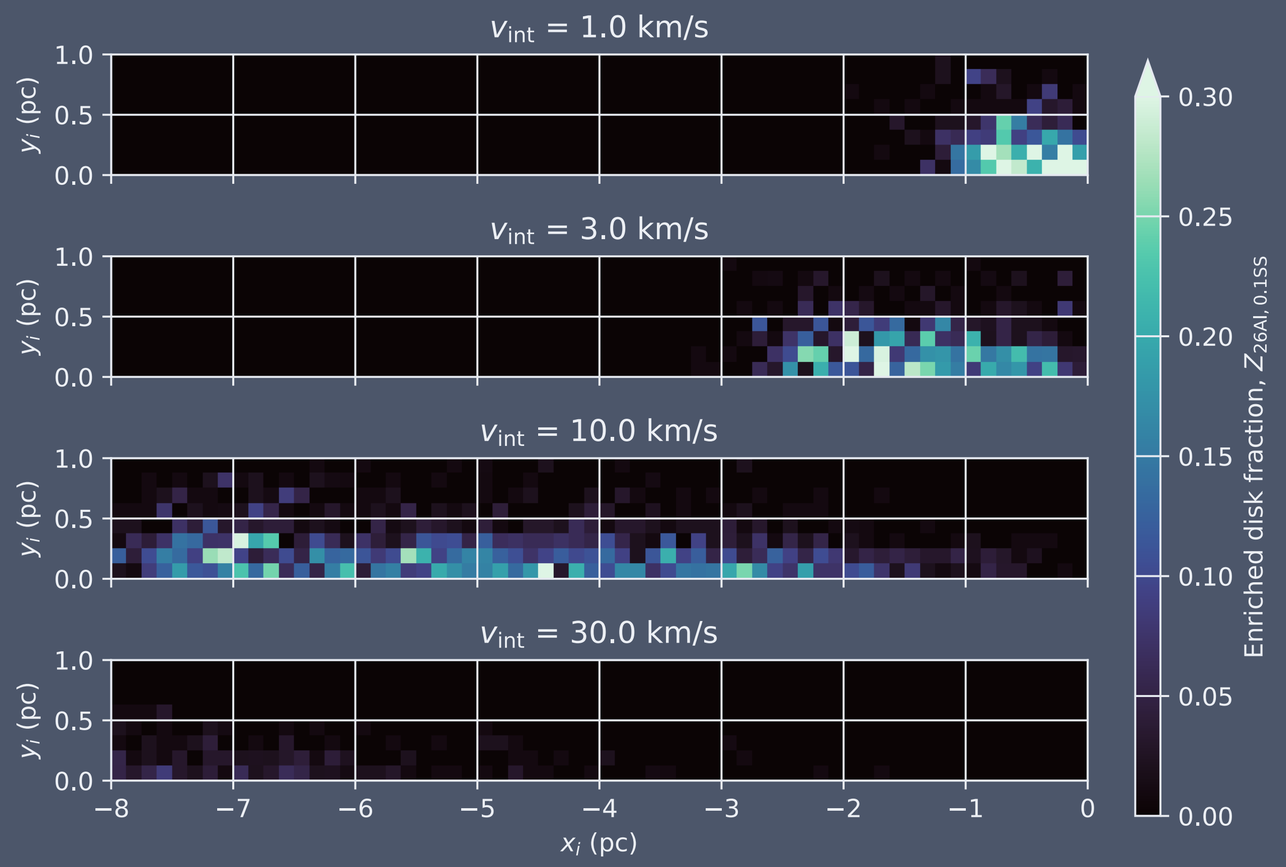

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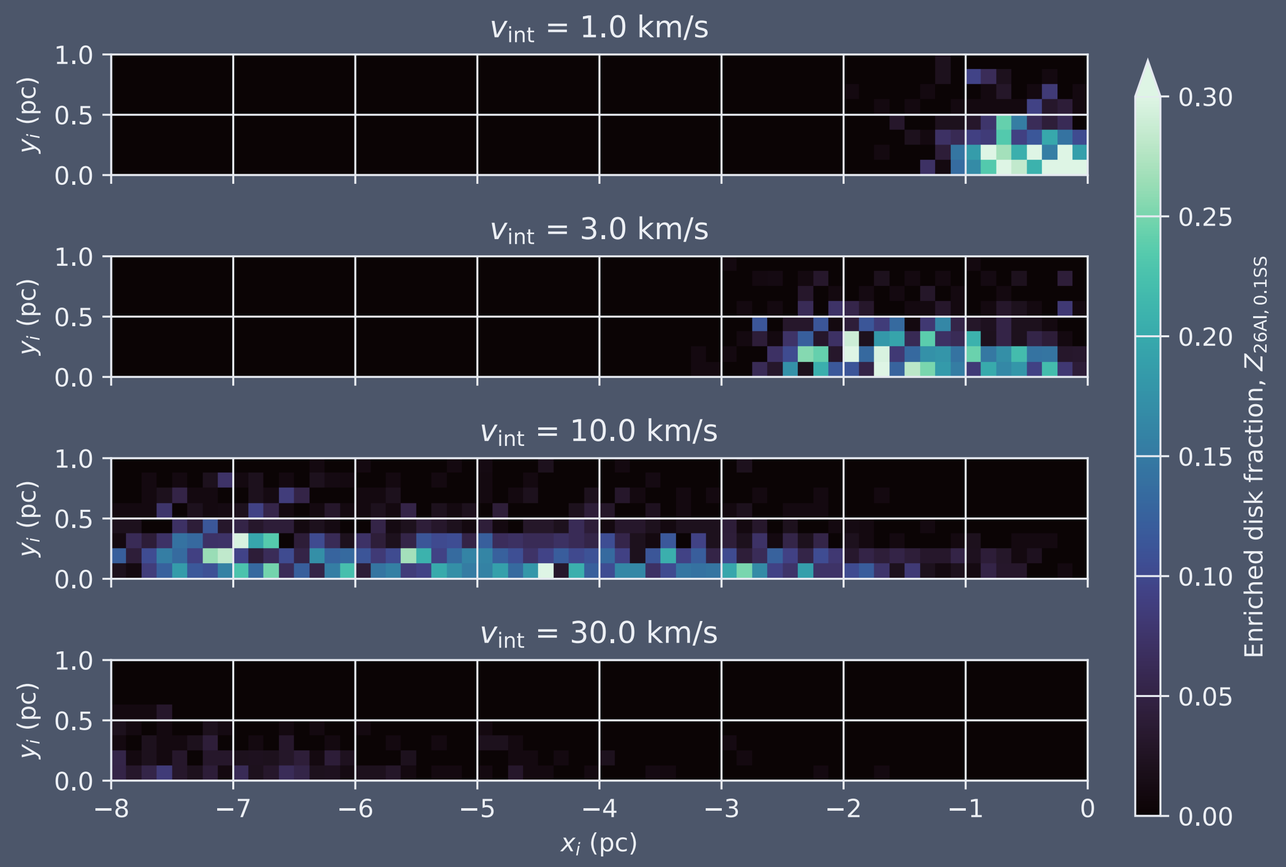

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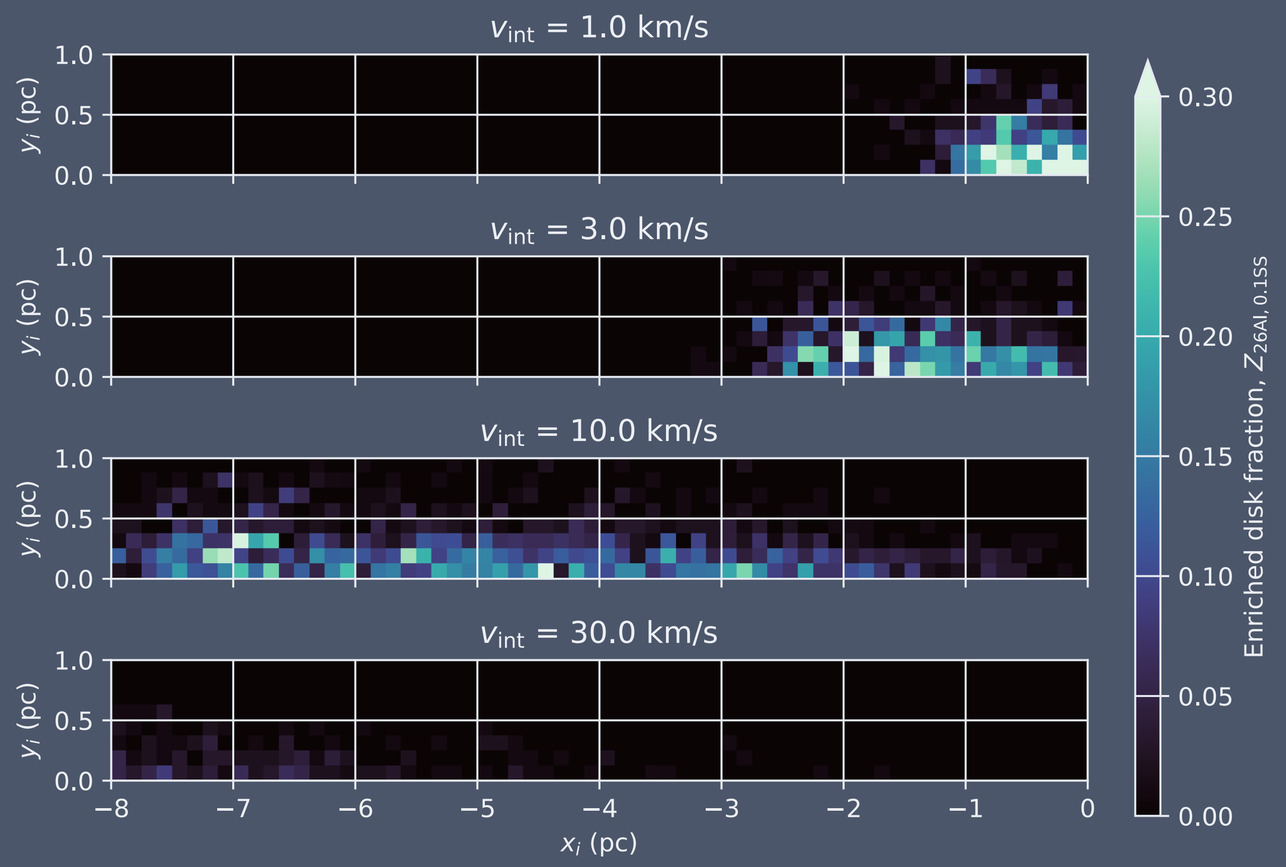

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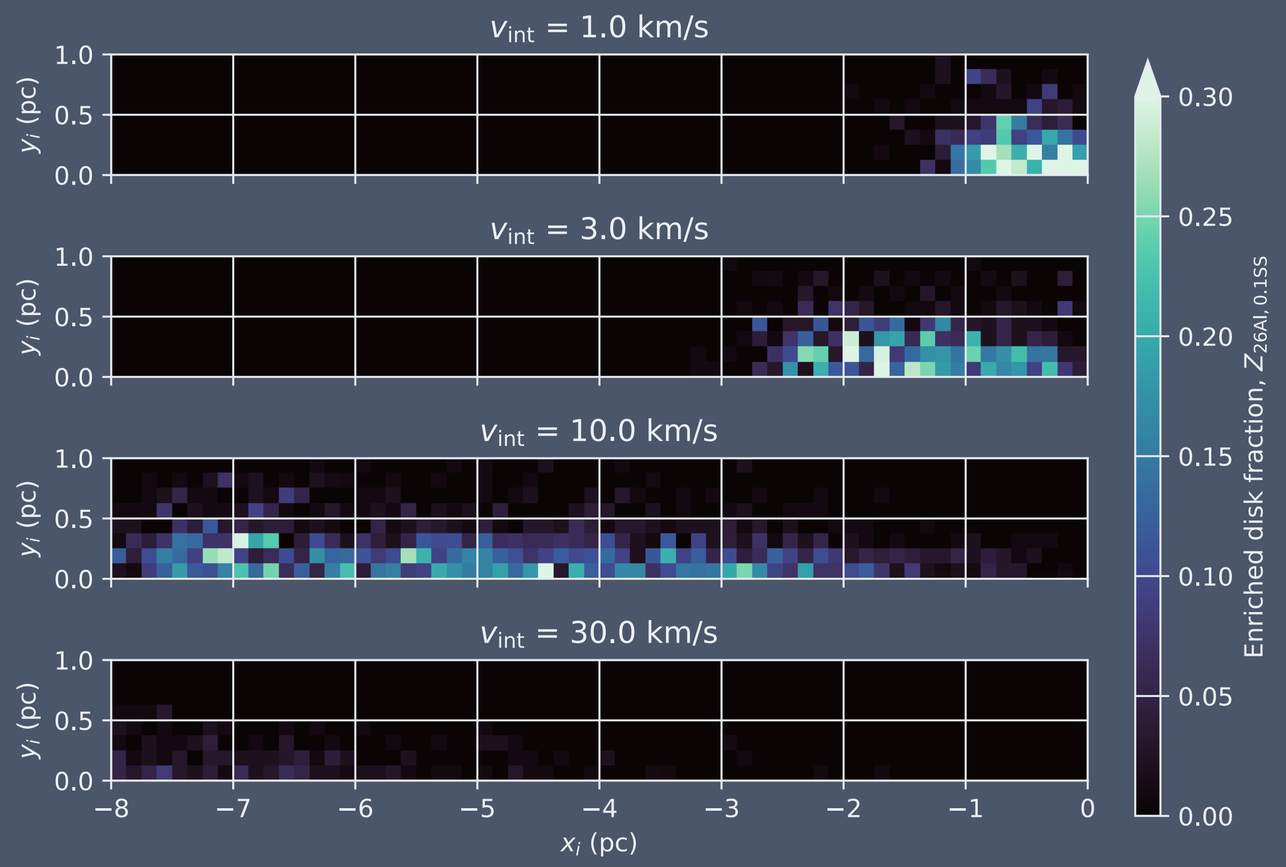

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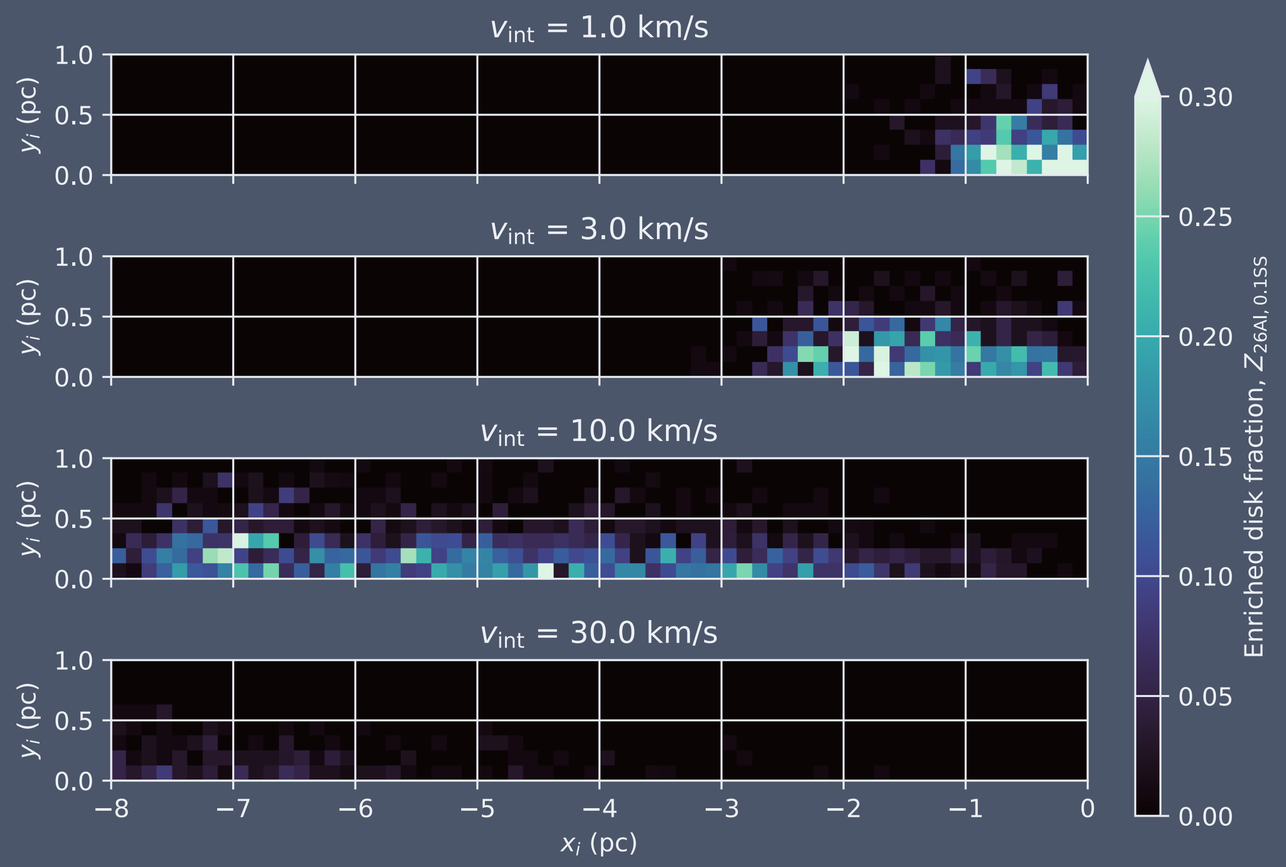

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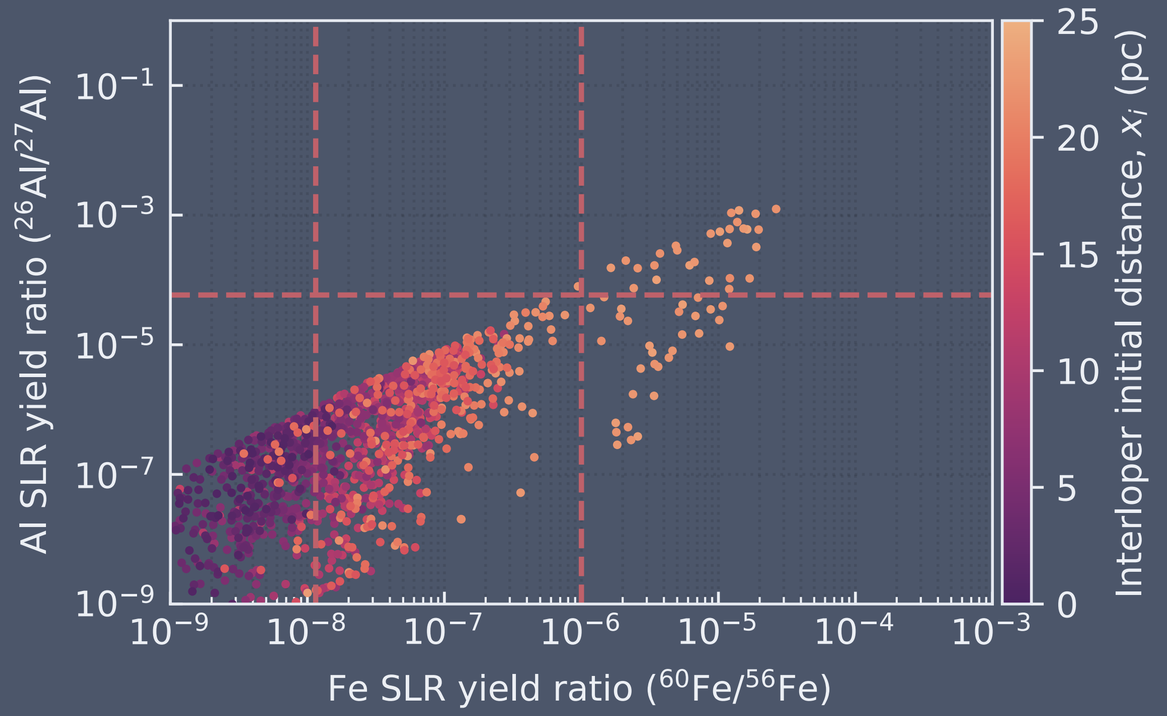

Planetesimal heating by SLRs

More motivations

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How much does the SLR ^{60}Fe influence heating and desiccation?

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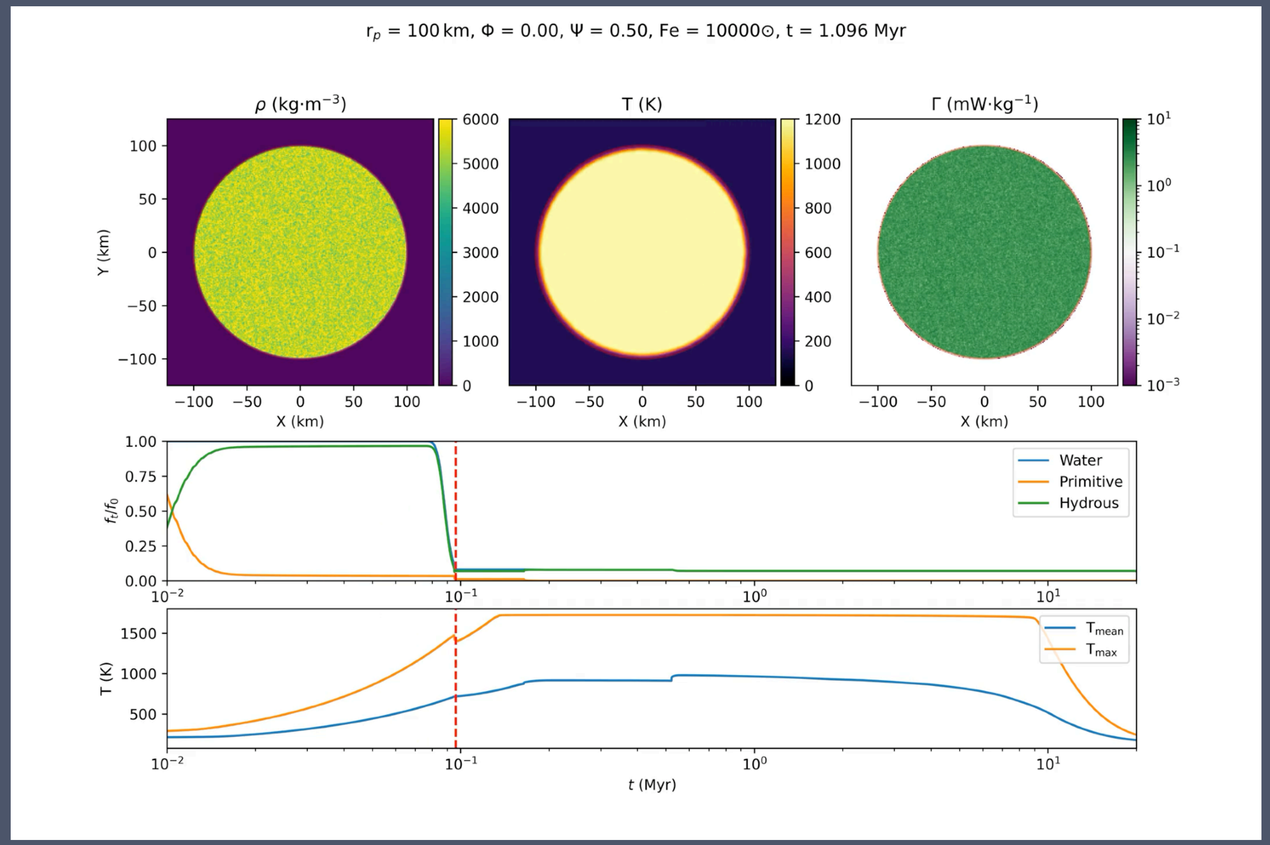
How much ^{60}Fe needed to get desiccation?

Melting rocks

Eatson, Parker, Lichtenberg & Gerya 2024



[1] Gerya (2019), github.com/FormingWorlds/i2elvis_planet

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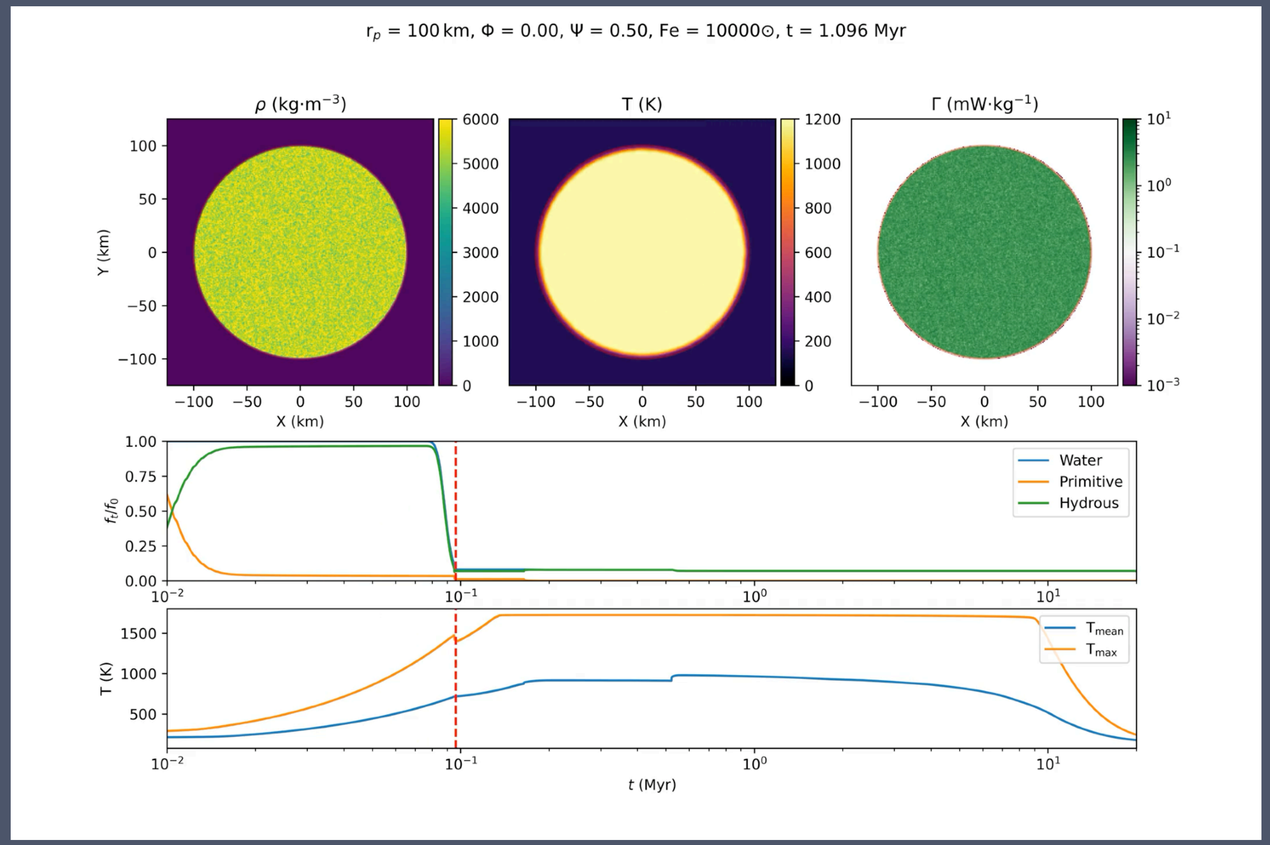

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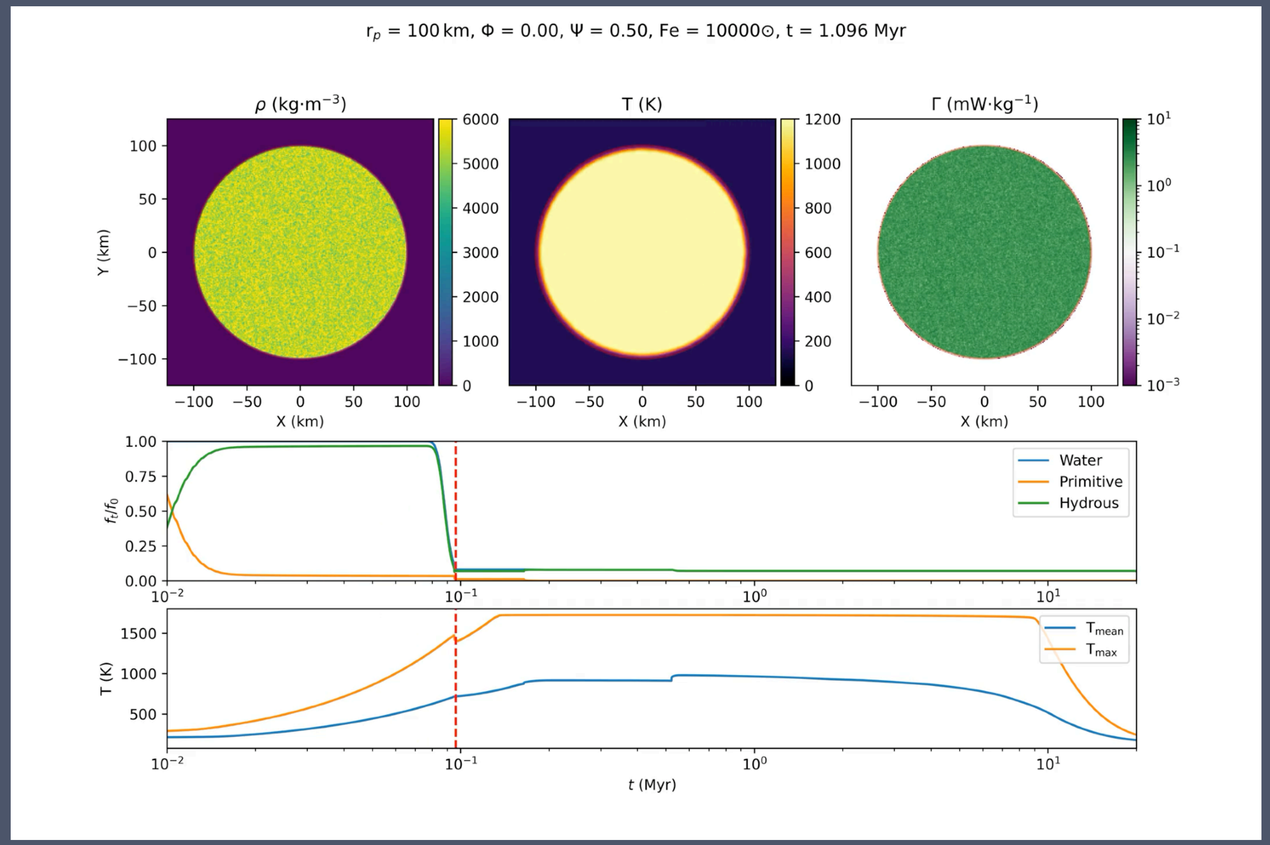

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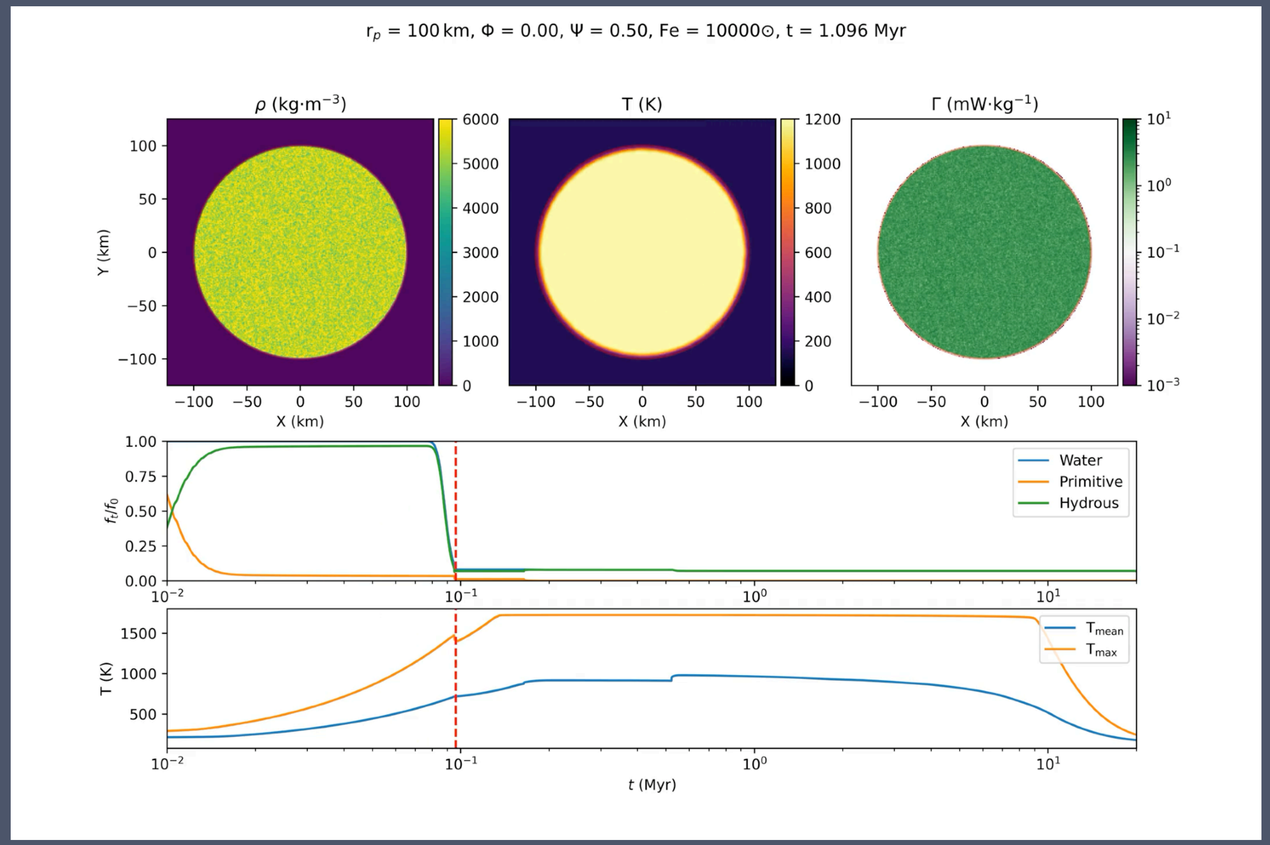

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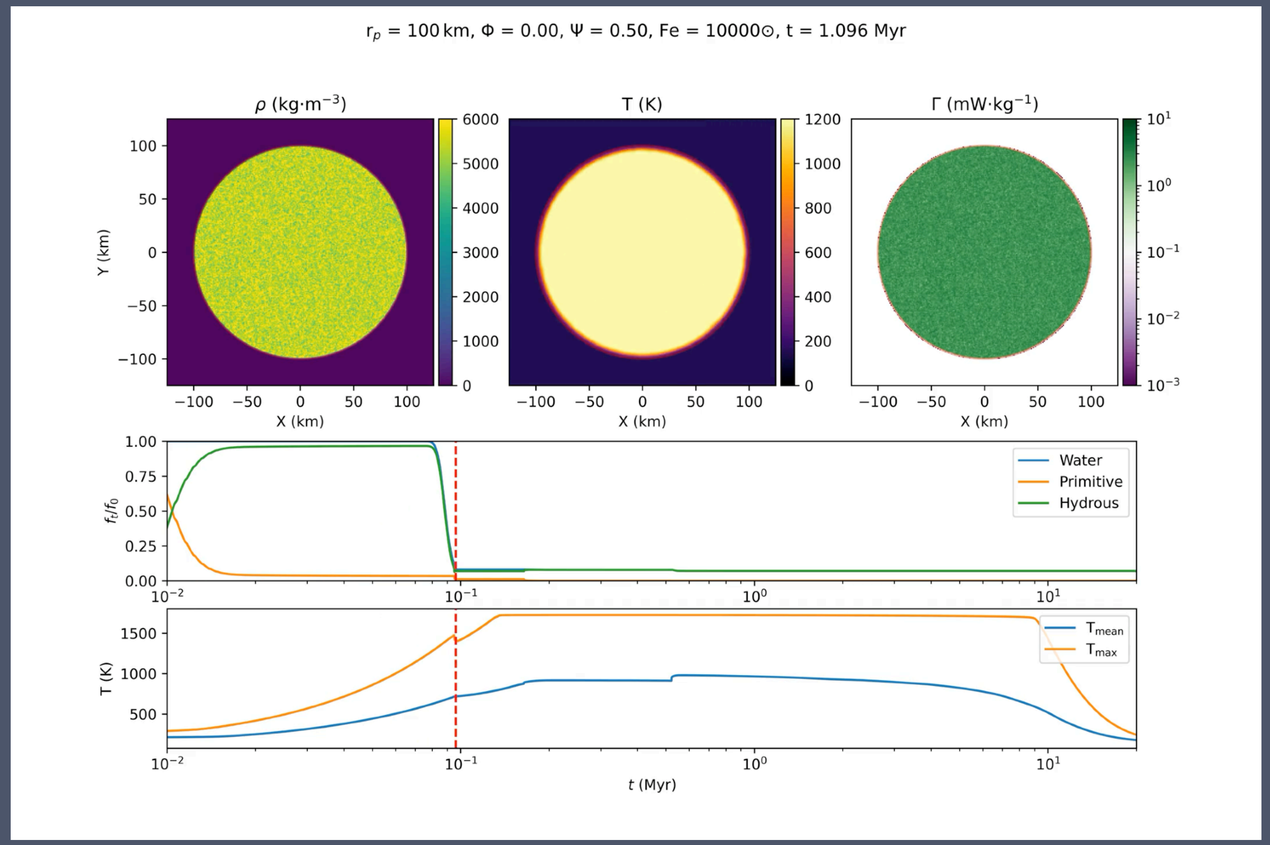

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Drying out rocks

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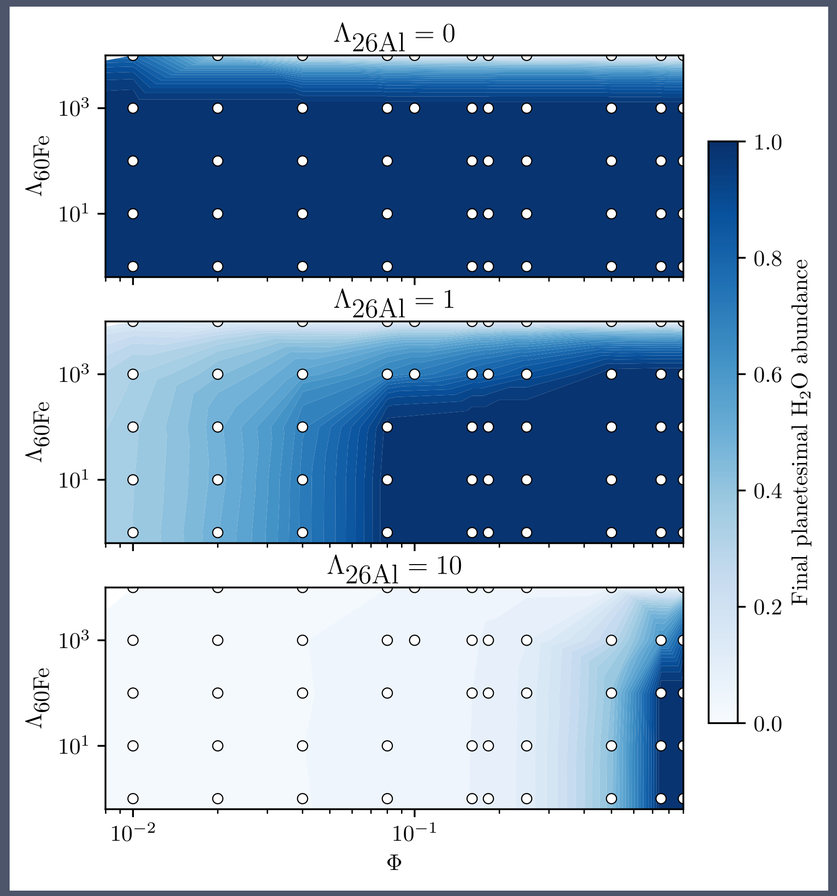

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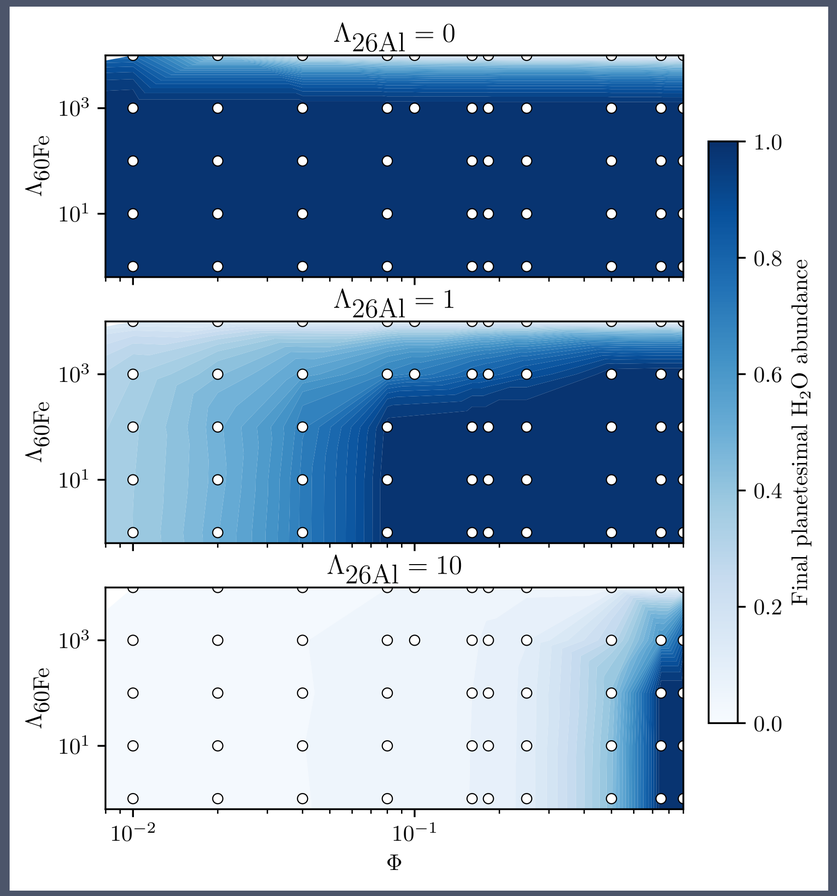

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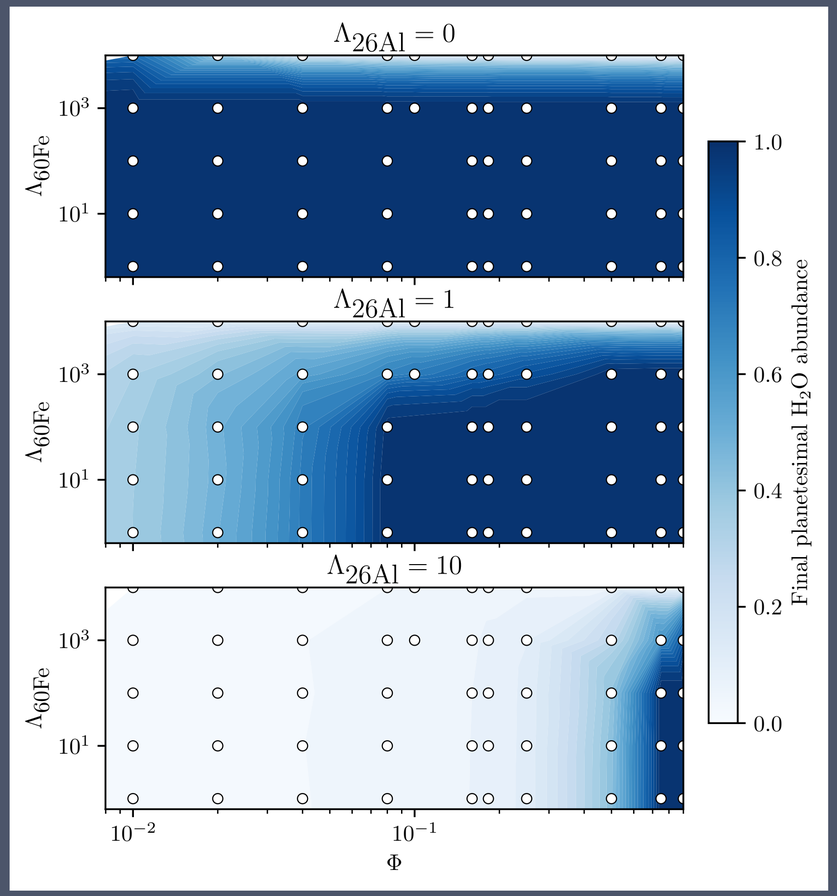

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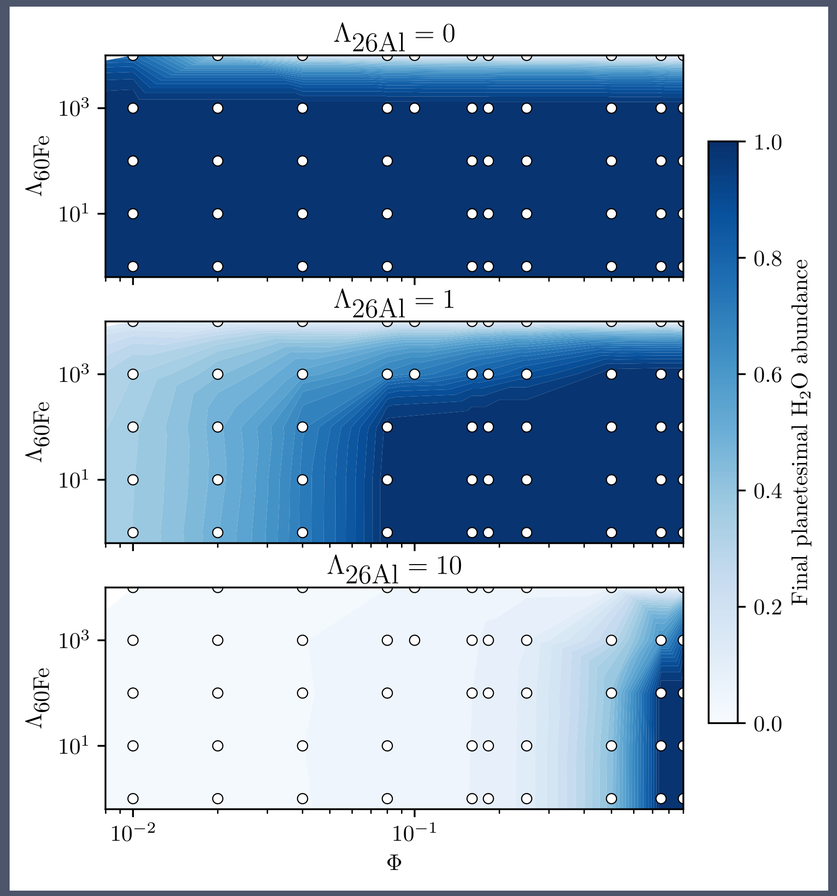

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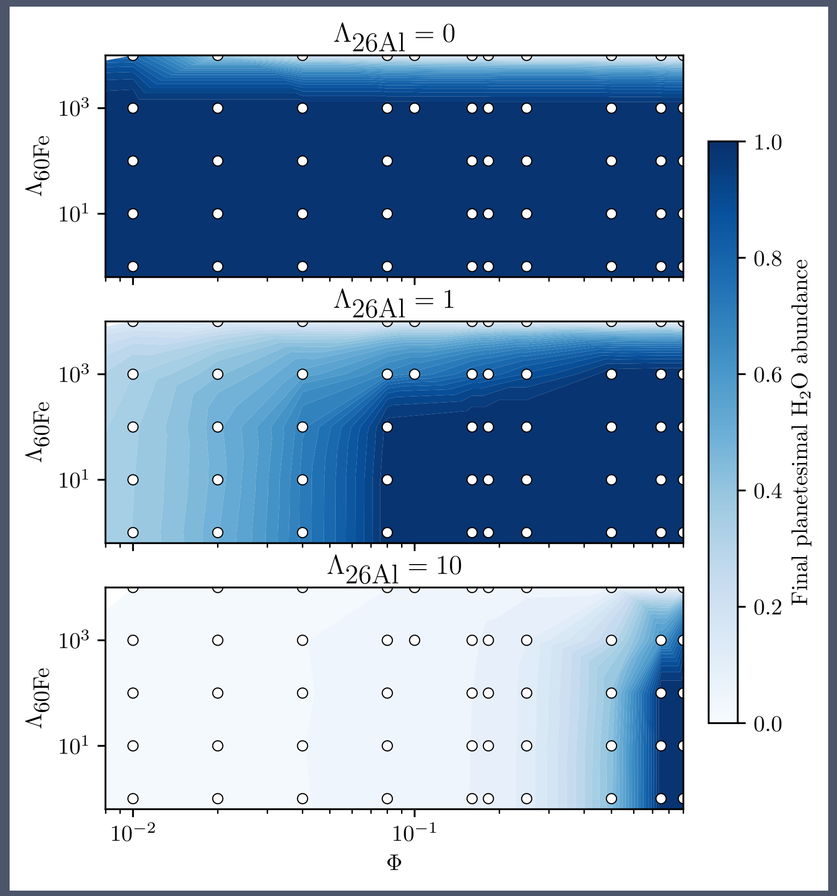

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Probably gone on too long

Wrap it up!

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- Decay heating mainly through ^{26}Al, sims show ^{60}Fe needs high enrichment.
- *N*-body sims show that massive star ^{60}Fe enrichment is not sufficient for this level of enrichment.
- Enrichment through AGB interlopers provides an alternate, gentler route to disk enrichment for ^{26}Al and ^{60}Fe.

Questions?

Slide deck and poster here!

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